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Report on One-Day Workshop on “From Idea to Publication: Writing Research Papers with LaTeX”

Introduction

The Department of Computer Science and Engineering (CSE) at ATME College of Engineering, Mysuru, organized an intensive one-day workshop titled **“From Idea to Publication: Writing Research Papers with LaTeX”** on **20 November 2025** at the **Seminar Hall, 401**. The workshop was specifically designed for **5th semester CSE students**, who are at a crucial stage in their academic journey, where research exposure, academic writing, and publication skills become highly relevant.

The key aim of the workshop was to enhance students’ understanding of the **research lifecycle**, introduce them to **LaTeX**, and equip them with the skills required to prepare high-quality technical documentation. Research publication is a vital component of academic growth, and exposure to LaTeX offers students a competitive edge in preparing professional manuscripts for conferences, journals, project reports, and technical presentations.

The event was coordinated by the faculty members of the CSE department, under the guidance of the Principal, **Dr. L Basavaraj**, with active support from **Dr. Puttegowda D**, HOD, Department of CSE.

Objectives of the Workshop

The workshop had the following major objectives:

Understanding Research Fundamentals

- To familiarize students with the concept of research, its purpose, and its importance in academic and industrial settings.
- To help them identify research problems and areas of interest.

Understanding the Research-to-Publication Lifecycle

- To provide a clear view of the journey from idea generation to the submission of a research paper.
- To introduce students to ethical research practices, plagiarism awareness, and citation styles.

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Gaining the Ability to Write Research Papers Professionally

- To help students understand the structure of scientific papers: Title, Abstract, Introduction, Literature Review, Methodology, Results, Discussion, Conclusion, and References.
- To train them in IEEE/ACM formatting standards.

Mastering LaTeX for Academic Writing

- To introduce students to LaTeX, its advantages, and popular editors like Overleaf.
- To enable students to create well-structured, professionally formatted documents.

Hands-on Practice

- To provide practical experience in writing LaTeX code.
- To encourage students to prepare tables, figures, equations, citations, and references using LaTeX.

The workshop was delivered by **Dr. Deepak N. A**, Associate Professor, Department of Computer Science and Engineering, RV Institute of Technology and Management, Bengaluru.

Dr. Deepak N. A is well known for his expertise in:

- Research paper writing and publication,
- LaTeX typesetting and document preparation,
- IEEE/ACM formatting guidelines,
- Technical writing for academic and industry applications.

His session brought real-world insights and practical tips that greatly benefited the participants.

Workshop Overview and Session Details

The workshop was conducted in multiple structured sessions that combined **conceptual understanding, technical demonstrations, and hands-on practice.**

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Session I: Introduction to Research and Publication Workflow

Dr. Deepak began the workshop with an overview of **what constitutes research**. He explained how research problems are identified by analyzing existing gaps in the literature.

Key topics covered:

- Definition and scope of research
- Difference between academic, scholarly, and industrial research
- Components of a research question
- Importance of literature review
- Structure of a research paper
- Role of journals, conferences, and publishers
- Peer-review process
- Academic integrity and plagiarism

The resource person also highlighted tools like **Google Scholar**, **IEEE Xplore**, **ACM Digital Library**, and **Scopus** for academic searches.

Session II: Introduction to LaTeX

The second session focused on LaTeX basics. Students were introduced to:

- Overview of LaTeX and its advantages compared to MS Word
- Installation and use of LaTeX editors
- Introduction to Overleaf (cloud-based LaTeX editor)
- Structure of a .tex file
- Document classes: article, report, IEEEtran, etc.
- Role of packages such as amsmath, graphicx, geometry, and biblatex
- Sectioning commands: \section, \subsection, \subsubsection
- Creating lists, tables, figures, and mathematical equations

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Students understood why LaTeX is the preferred writing tool in scientific research, particularly for technical subjects involving equations and formatting.

Session III: Hands-on Training

This was the most interactive session where students practiced LaTeX coding in real-time.

Activities included:

- Creating Title, Abstract, and Introduction sections
- Inserting tables using tabular environment
- Adding figures and captions using the graphicx package
- Writing mathematical equations using equation and align environments
- Including inline and block mathematics
- Generating citations and references using BibTeX
- Understanding how compilation works (PDFLaTeX, XeLaTeX)

Students also created a **sample IEEE-formatted LaTeX file**, guided step-by-step by the resource person.

Session IV: IEEE Format and Publication Standards

The final session covered advanced formatting based on IEEE standards.

Topics included:

- IEEE two-column layout
- Creating the title block and author information
- Formatting figure/table placement
- Using IEEE citation and reference styles
- Organizing keywords, abstract, and main content
- Understanding camera-ready paper submission

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- Common mistakes in IEEE formatting

The resource person also shared examples of published IEEE papers to help students understand real-world applications.

Student Participation and Engagement

The workshop was attended by enthusiastic 5th semester CSE students. Students interacted actively, raised questions, and tried multiple LaTeX templates during the hands-on sessions. Many students said it was their first experience with LaTeX, and they found it highly useful for upcoming:

- Mini-project reports
- Project documentation
- Technical presentations
- Journal/Conference publications

The learning environment was highly engaging, with discussions on real research trends, citation tools, and best practices.

Outcomes of the Workshop

By the end of the workshop, students gained:

Technical Outcomes

- Ability to use LaTeX for writing technical documents
- Knowledge of BibTeX for automated referencing
- Skills to format documents using IEEE templates
- Understanding of compiling and debugging LaTeX code

Academic Outcomes

- Improved understanding of research methodology
- Enhanced capability to structure research papers
- Awareness of publication standards

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- Confidence to submit papers to conferences

Professional Outcomes

- Better preparedness for internships and project reviews
- Strong foundation for research-oriented careers
- Competitive advantage in documentation skills

Feedback from Participants

Participants expressed that the workshop was:

- Highly informative
- Practical and easy to follow
- Well-structured with hands-on exercises
- Helpful for future academic and project work

Many students conveyed their intention to adopt LaTeX for all future documentation requirements.

Conclusion

The one-day workshop on “**From Idea to Publication: Writing Research Papers with LaTeX**” was a resounding success and met all its intended objectives. It significantly enhanced students’ understanding of research writing, LaTeX typesetting, and IEEE formatting standards.

The Department of CSE expressed its gratitude to:

- The resource person, **Dr. Deepak N. A,**
- The Principal, **Dr. I. Basavaraj,**
- The Convenor, **Dr. Puttegowda D,**
- Event Coordinator, **Bindushree V,**
- Faculty members and participants

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for their effort, involvement, and support in making the workshop engaging and impactful.

PO Addressed	PSO Addressed
PO1, PO2, PO3, PO5, PO9, PO10	PSO1

Glimpse of the workshop



Welcoming the guest

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HOD addressing the audience



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